AI strength more than offsetting smartphone/PC semis weakness; OW

April 16, 2026 05:30 PM GMT

WHAT'S CHANGED

	From	To
TSMC (2330.TW)		
Price Target	NT$2,288.00	**NT$2,588.00**

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We raise our revenue and margin assumptions after TSMC's strong print today.

2026 revenue raised to above 30% Y/Y: TSMC guided 2Q26 revenue to be up 10% Q/Q, beating our and the Street's 5%-10% Q/Q. Despite PC/smartphone weakness and some macro uncertainty, AI demand remains very strong. TSMC also raised full-year growth to 30%+, vs. "close to 30%" Y/Y. We therefore raise our 2026 revenue forecast to up 36% Y/Y in USD.

Gross margin beat leads to our upwards earnings revision: 1Q26 GM was 66.2%, above MSe 64.2%. The 2Q26 GM midpoint is 66.5%, also above MSe's 64.4%. Better productivity and cost savings drove the improvement; we also believe a better product mix helped (more HPC vs. smartphone/consumer). We expect 2H26 GM to hold near 66%, offset by some 2nm dilution. We raise 2026-2028e EPS by 8%/11%/13%, respectively, mainly on margin upside.

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2026 capex raise: TSMC lifted 2026 capex to the upper end of US$52-US$56bn. AI-related 3nm demand (including HBM base die) is very strong. Combining our recent equipment supply chain checks (see our Foundry Floorplan on 3nm and EUV bookings), we introduce 3-year capex of US$200bn: US$55bn in 2026 (low-20 EUV units), US$70bn in 2027 (c.40 units), and US$75bn in 2028 (low-40 units).

Addressing competition (or collaboration) from Intel's EMIB: We asked about competition from Intel (covered by Joseph Moore) in advanced packaging for AI chips via its EMIB technology – TSMC management said the company welcomes competition and is fine to open the compute die to third-party foundries for packaging to enlarge AI semi TAM. That said, TSMC does not intend to forgo value and will keep offering CoWoS-based solutions at commercially reasonable costs for large-reticle designs. Long term, its CoPoS, SoIC, and SoW portfolio will together address customers' larger-reticle designs.

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Mature node foundry: TSMC will take a few years to wind down Fab 2 (6 *inch*) and Fab 5 (GaN) to optimize land and engineering resources. That said, TSMC's capacity support to mature node customers will not be compromised. In fact, TSMC's other mature node fabs are fully utilized. We thus keep UW on UMC.

Stay OW and raise PT to NT$2,588: Results and guidance fall in our CDI Scenario 1 Bull Case; we expect the stock should be up 3-5% in the coming trading days.

Key points from 1Q26 earnings call

See also 1Q26 earnings review and 2Q26 guidance review .

- **2Q26 revenue guidance beat:** Management guided for 2Q26 revenue to be up 10% Q/Q, better than our and consensus expectations of up 5-10% Q/Q.
- **2026 revenue revised up to 30%+ Y/Y:** This is similar to our updated preview of +35% Y/Y. Amid the weakness in PC/smartphone and macro uncertainty, AI-related demand remains solid.

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- **Gross margin beat:** 1Q26 gross margin reached 66.2%, vs. MSe 64.2%. The mid-point of 2Q26 guidance is 66.5%, vs MSe 64.4%. Chemical and energy costs may impact profitability; however, operations shouldn't be disrupted.
- **2026 capex raised:** TSMC updated its 2026 capex to the upper end of its US$52-56bn guidance given solid 3nm demand for AI-related chips (including HBM base die). While it did not provide a 3-year capex guidance of US$200bn that we had predicted, management expects capex to step up in the coming years and will work closely with equipment vendors (e.g., ASML) to ensure tool supply.
- **Wafer foundry competition:** There is no shortcut for new entrants such as TeraFab. Leading-edge foundry capability requires years to develop, with new fabs typically taking 2-3 years to build.
- **Intel EMIB competition:** TSMC welcomes competition and is okay to open the compute die to 3rd-party foundries for the packaging to enlarge the AI semiconductor TAM. That said, TSMC does not intend to forgo value and will continue to offer CoWoS-based solutions at commercially reasonable costs for large-reticle chip designs.

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In accordance with our Catalyst Driven Idea Scenario 1 bull case, we expect the stock should be up 3-5% in the coming trading days.

Exhibit 1: TSMC: CDI Table

Details of the scenario	Scenario 1	Scenario 2	Scenario 3
PROBABILITIES	25%	60%	15%
2Q26 USD revenue growth	>10% Q/Q	5-10% Q/Q	<5% Q/Q
2Q26 gross margin	>65%	64-65%	<63%
2026 full-year capex in USD	Raise to US$56-60bn	Higher end of US$52-56bn	Lower end of US$52-56bn
Impact from geopolitical tensions	None	Mild	Severe
Impact from non-AI demand	None	Mild	Severe
POTENTIAL CHANGE IN STOCK PRICE (%)	UP 3-5%	UP 1-2%	DOWN 6-8%

Source: Morgan Stanley Research

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Key Q&A from the earnings call

1. Geopolitical impact: Fab operation and margin impact from the recent conflict.

Chemical and energy costs may indeed have some impact to profitability, but there shouldn't be operational disruption. TSMC also reiterated that it has multiple suppliers for specialty chemicals, including helium and hydrogen.

2. New foundry competition: Last time we asked about Intel's Foundry competition; this time Tesla's TeraFab took over the topic of foundry competition.

On the new entrants, such as TeraFab, TSMC believes there is no shortcut, as it takes years for a leading-edge foundry service to mature, including 2-3 years for new fab construction, and then few years of customer design in and production ramp up. TSMC is also expanding overseas capacity (including the US fabs) to address customer demand.

3. Non-AI demand outlook: **Impact to TSMC should be minimal.**

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When asked about the non-AI demand impact, TSMC management acknowledged that consumer demand and price-sensitive segments were indeed a bit softer. However, the company still raised full-year revenue guidance to 30%+Y/Y, suggesting AI demand should more than cover potential downside from consumer.

4. Three-year capex outlook: Management didn't share clear numbers for capex beyond 2026.

Although management didn't share the 3-year capex number as we'd initially expected, they did indicate capex will step up in the coming years due to very strong customer demand, not competition. Meanwhile, TSMC is working with equipment venders (e.g., ASML) closely to ensure tool supply.

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We raise our 2026-28 capex assumption to US$200bn with 2026 revenue up mid-high 30% Y/Y

A month ago we published a report Greater China Semiconductors: The Foundry Floorplan: Raise TSMC's 2027e and 2028e Capex (9 Mar 2026) to address that we had been seeing TSMC aggressively pull in capacity build across the world. Our assumption of TSMC 3-year capex was then close to US$190bn in total.

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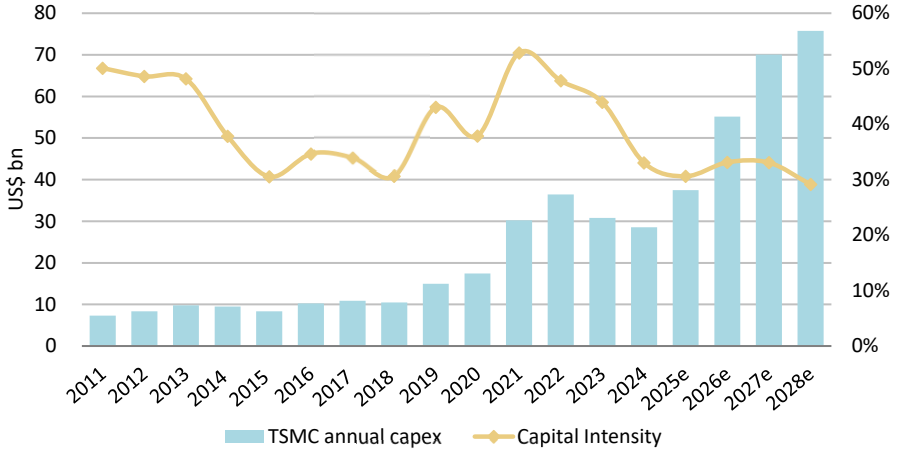
Although management did not share the 3-year capex estimate during 1Q26 earnings call, they did share that they see very strong demand for 3nm process and have built additional capacity in Tainan Science Park, Arizona Fab 2, and Kumamoto fab 2, with ongoing capacity conversion from N5 to N3.

Combining our own industry checks, TSMC's aggressive pull in of EUV scanners in 2027-28, as well as our own bottom-up analysis of TSMC 3nm/2nm demand, which shows UTR is likely to reach >100% in the following years, we raise our total 3-year capex forecast to US$200bn. 2026e capex is raised from US$54bn to US$55bn, followed by US$70bn and US$75bn in 2027 and 2028.

In addition, we also raise our 2026e full-year revenue forecast from low-30% Y/Y growth to mid-high 30% Y/Y in USD, following the strong revenue beat in both 1Q26 and 2Q26 (guidance beat).

Exhibit 2: TSMC's capex intensity should drop sharply with a large revenue ramp up



Source: Company data, Morgan Stanley Research (e) estimates

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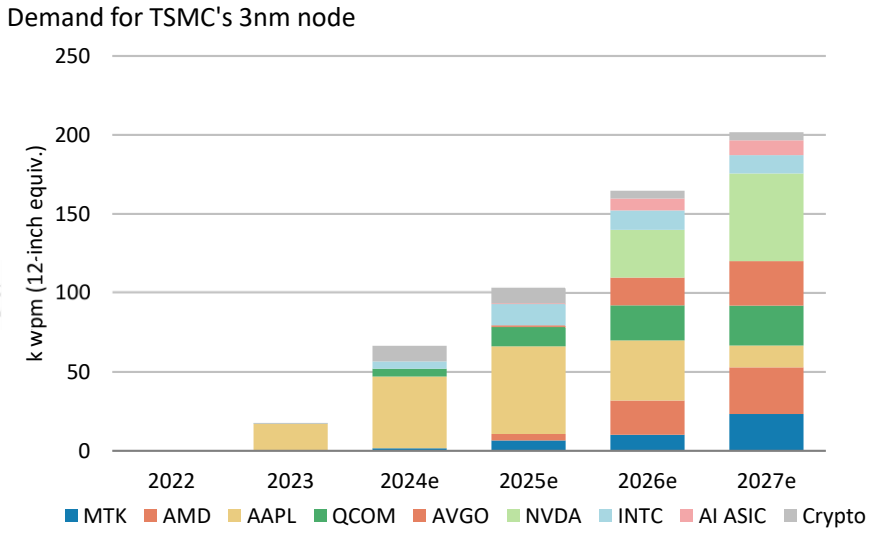
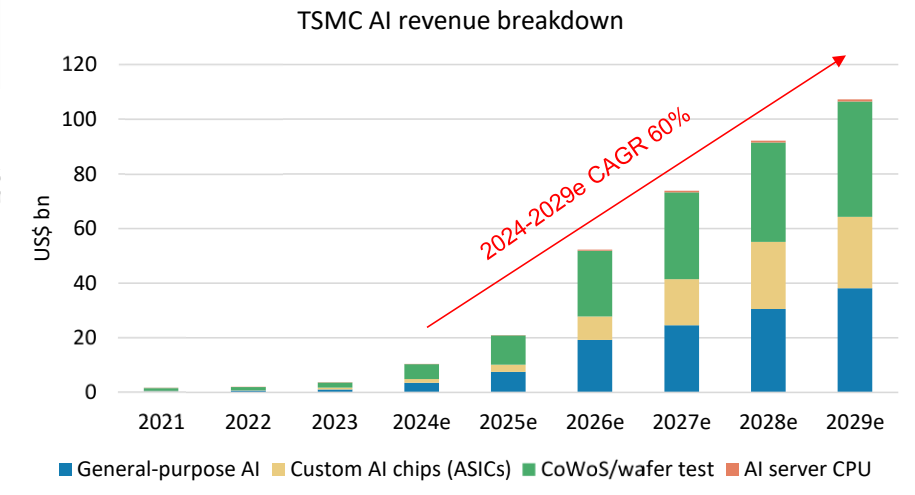
Exhibit 3: Wafer demand for TSMC 3nm



Source: Morgan Stanley Research (e) estimates

Exhibit 4: TSMC's AI revenue could achieve a 55%-60% CAGR during 2024 to 2029



Source: Company data, Morgan Stanley Research (e) estimates

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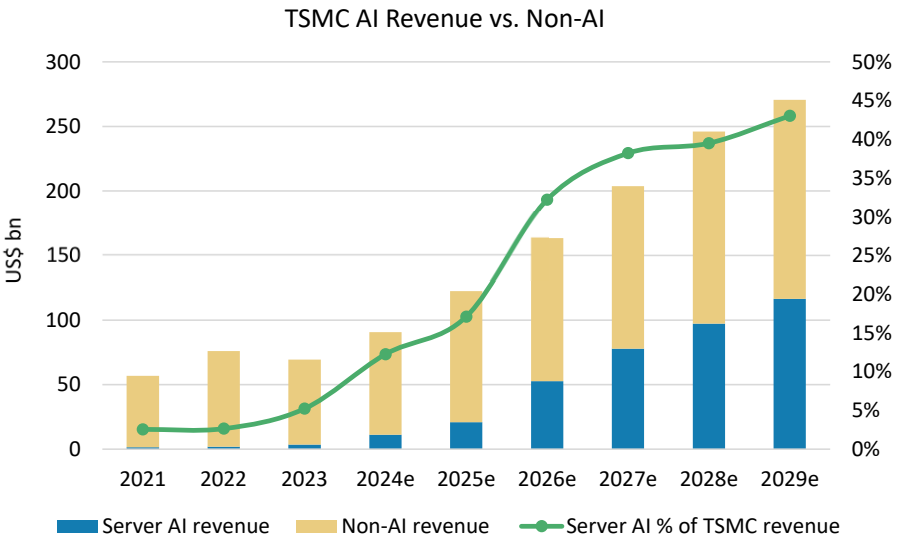
Exhibit 5: TSMC mentioned that AI revenue accounted for a high-teens share of 2025 total revenue, and we expect it to reach 40% by 2027



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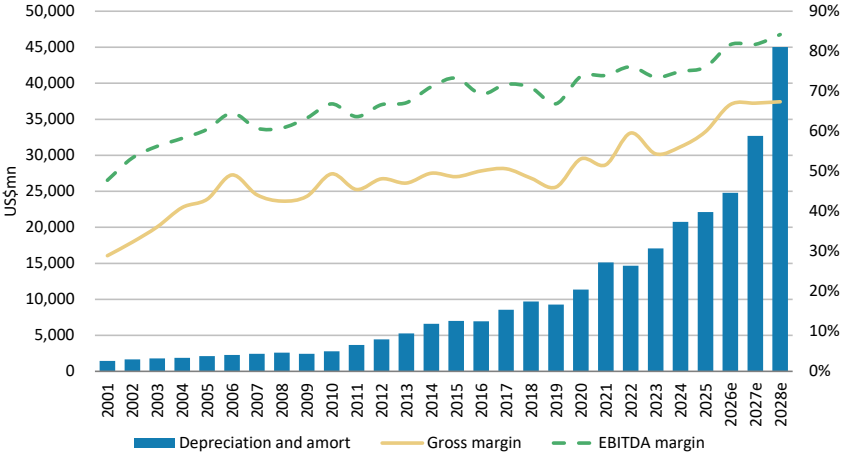
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 6: We think TSMC can easily achieve its >56% gross margin goal even with increased capex



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Source: Company data, Morgan Stanley Research (e) estimates

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1Q26 earnings review

TSMC's 1Q26 gross margin was 66.2%, beating both our estimate and consensus by around 2ppts, as well as its guidance of 63%-65%. The margin beat was mostly from a higher utilization rate, as well as productivity improvement. Looking into the 1Q26 revenue breakdown, we are seeing 5nm demand sustain strong momentum from 4Q25, contributing around 36% of the total revenue, suggesting strong demand from major AI product and peripheral demand.

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Exhibit 7: 1Q26 earnings review

(NT$ mn)	1Q26				
	Actual	Q/Q	Y/Y	MS Est.	Consensus
Financials					
Revenue	1,134,103	8.4%	35.1%	1,134,103	1,125,493
Opex	(92,329)	6.0%	7.0%	(101,023)	(101,246)
EPS (NT$)	22.08	13.2%	58.3%	21.43	20.89
Ratios					
GM (%)	66.2%	392bps	746bps	64.2%	64.3%
Opex (%)	8.1%	-18bps	-214bps	8.9%	9.0%
OPM (%)	58.1%	410bps	960bps	55.3%	55.3%

Source: Company Data, FactSet, Morgan Stanley Research estimates

2Q26 guidance review

Looking to 2Q26, the company guided for USD revenue to be up 10% Q/Q at the midpoint, which is higher than our estimate of 5-10% Q/Q growth. In terms of margin, TSMC guides for 2Q26 GM to be 66.5% at the midpoint, which also strongly beat our estimate and consensus by around 2ppts. However, despite the strong beat in 1H26 margin, we don't expect 2H26 margins to keep going up on H/H basis. 2H26 margin is likely to stay at 66.5%-67.0% given the dilution from the 2nm process, for which the company has already guided for 2-3% margin dilution from 2nm process to the overall corporate margin.

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Exhibit 8: 2Q26 Guidance Review

(NT$ mn)	2Q26				
	Guidance	MS Est.	Q/Q	Y/Y	Consensus
Financials					
Revenue	US$39.0bn to US$40.2bn, +10% Q/Q increase at mid-point	1,220,992	7.7%	30.8%	1,206,441
Opex		(105,119)	13.9%	25.2%	(107,558)
EPS (NT$)		22.67	2.7%	47.6%	22.10
Ratios					
GM (%)	65.5% - 67.5%	64.4%	-181bps	582bps	64.2%
Opex (%)		8.6%	47bps	-38bps	8.9%
OPM (%)	56.5% - 58.5%	55.8%	-228bps	620bps	55.3%

Source: Company Data, FactSet, Morgan Stanley Research estimates

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EPS estimate revisions

We raise our EPS estimates 8% for 2026, 11% for 2027, and 13% for 2028: We raised our revenue forecasts by 3-6% as the company's ability to push the UTR further has been better than our expectation. However, our upward earnings revisions are are mostly due to margin improvement driven by higher UTR and ongoing productivity improvement from the company.

Exhibit 9: TSMC: Estimate revisions

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(NT$ mn)	New 2026e	Old 2026e	Diff.%	New 2027e	Old 2027e	Diff.%	New 2028e	Old 2028e	Diff.%
Net sales	5,213,620	5,073,332	*3%*	6,566,633	6,330,127	*4%*	8,060,022	7,626,120	*6%*
Gross profit	3,476,058	3,262,561	*7%*	4,401,351	4,014,283	*10%*	5,430,454	4,840,389	*12%*
Operating profit	3,076,809	2,827,031	*9%*	3,850,597	3,470,515	*11%*	4,772,653	4,195,105	*14%*
Pretax income	3,185,165	2,933,060	*9%*	3,956,627	3,576,545	*11%*	4,874,683	4,297,135	*13%*
Net income	2,679,677	2,478,938	*8%*	3,344,231	3,022,546	*11%*	4,107,817	3,621,179	*13%*
Diluted EPS	**103.34**	**95.60**	***8%***	**128.97**	**116.57**	***11%***	**158.42**	**139.65**	***13%***
Margins									
Gross margin	66.7%	64.3%		67.0%	63.4%		67.4%	63.5%	
Operating margin	59.0%	55.7%		58.6%	54.8%		59.2%	55.0%	
Pretax margin	61.1%	57.8%		60.3%	56.5%		60.5%	56.3%	
Net margin	51.4%	48.9%		50.9%	47.7%		51.0%	47.5%	

Source: Morgan Stanley Research (e) estimates

Valuation methodology

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We raise our target price from NT$2,288 to NT$2,588, the upward revision is due mostly to our earnings estimates change for 2026e to 2028e. We continue to derive our 12-month price target (base case scenario) using a residual income valuation methodology. All of our key assumptions remain unchanged, including a cost of equity of 9.2% (beta of 1.2, risk premium of 6.0%, risk-free rate of 2.0%), intermediate growth rate of 10.5%, and terminal growth rate of 4%.

Our Bull case rises from NT$2,760 to NT$3,120, and our Bear case rises from NT$1,260 to NT$1,425.

Exhibit 10: TSMC: Residual income model

	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	**7,569,957**	**10,162,145**	**13,310,459**	**16,215,507**	**19,425,586**	**22,972,722**	**26,892,308**	**31,223,451**	**36,009,363**	**41,297,796**	**47,141,515**	**53,598,824**
Net Profit	**2,679,677**	**3,344,231**	**4,107,817**	**4,539,138**	**5,015,747**	**5,542,401**	**6,124,353**	**6,767,410**	**7,477,988**	**8,263,177**	**9,130,810**	**10,089,546**
ROAE	**41.1%**	**37.7%**	**35.0%**	**30.7%**	**28.1%**	**26.1%**	**24.6%**	**23.3%**	**22.2%**	**21.4%**	**20.6%**	**20.0%**
Residual Income	1,743,551	2,158,915	2,621,930	2,867,972	3,072,169	3,291,557	3,529,465	3,788,969	4,073,116	4,385,048	4,728,086	5,105,801
Spread	**31.9%**	**28.5%**	**25.8%**	**21.5%**	**18.9%**	**16.9%**	**15.4%**	**14.1%**	**13.0%**	**12.2%**	**11.4%**	**10.8%**
Ending Equity Capital	7,569,957											
PV of Forecast Period	20,747,529											
PV of Continuing Value	38,783,309											
Equity Value	**67,100,795**											
No. of Shares	25,930											
Projected Price (NT$)	**2,588**											

Source: Company data, Morgan Stanley Research (e) estimates

Relative valuation

The stock is trading at 16.3x our 2027 EPS estimate, near its NTM average P/E of 16x since 2018. We find this level very attractive.

Exhibit 11: TSMC's one-year forward P/E: valuation does not appear stretched as a key AI semi enabler

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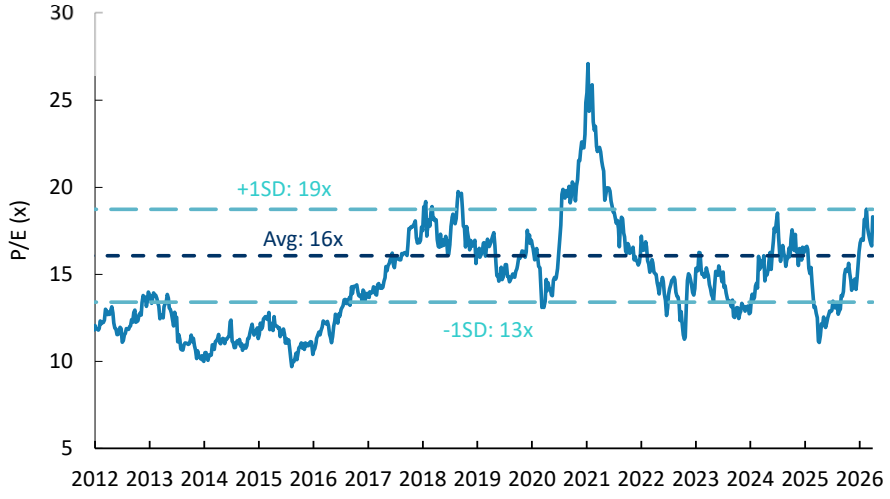

Source: Company data, FactSet, Morgan Stanley Research

Exhibit 12: TSMC P/B vs. ROE

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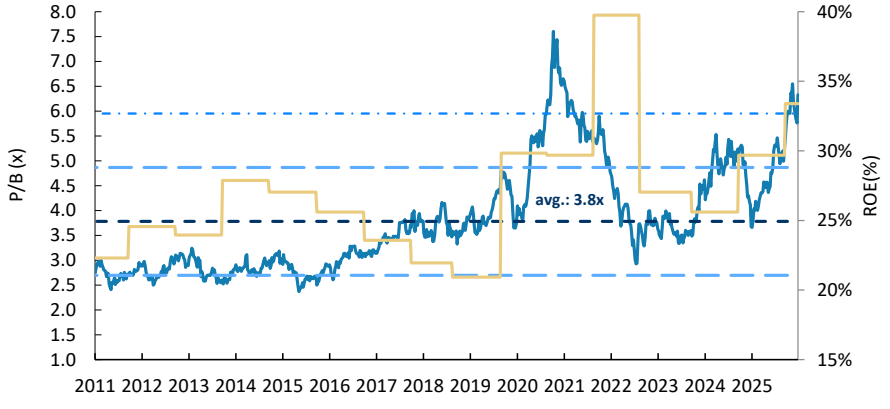

Source: Company data, Refinitiv, Morgan Stanley Research

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Morgan Stanley is acting as exclusive financial advisor to AP Grange Holdings, LLC ("AP Grange") in relation to a definitive agreement with Intel Corporation ("Intel"), for Intel to repurchase the 49% equity interest in the joint venture related to Intel's Fab 34 in Ireland not held by Intel, as announced on April 1, 2026. The transaction is subject to customary closing conditions. AP Grange has agreed to pay fees to Morgan Stanley for its financial advisory services. Please refer to the notes at the end of this report.

Exhibit 13: TSMC: Quarterly financials

(NT$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26e	2Q26e	3Q26e	4Q26e	2023	2024	2025	2026e	2027e	2028e
Total Revenues	**839,254**	**933,792**	**989,918**	**1,046,090**	**1,134,103**	**1,266,623**	**1,372,042**	**1,440,852**	**2,161,736**	**2,894,308**	**3,809,054**	**5,213,620**	**6,566,633**	**8,060,022**
Sequential Change	*-3.4%*	*11.3%*	*6.0%*	*5.7%*	*8.4%*	*11.7%*	*8.3%*	*5.0%*						
Change vs Year Ago	*41.6%*	*38.6%*	*30.3%*	*20.5%*	*35.1%*	*35.6%*	*38.6%*	*37.7%*	*-4.5%*	*33.9%*	*31.6%*	*36.9%*	*26.0%*	*22.7%*
Cost of Sales	**(345,859)**	**(386,423)**	**(401,375)**	**(394,103)**	**(382,808)**	**(418,298)**	**(458,622)**	**(477,833)**	**(986,625)**	**(1,269,954)**	**(1,527,760)**	**(1,737,562)**	**(2,165,282)**	**(2,629,568)**
Percent of Revenues	*41.2%*	*41.4%*	*40.5%*	*37.7%*	*33.8%*	*33.0%*	*33.4%*	*33.2%*	*45.6%*	*43.9%*	*40.1%*	*33.3%*	*33.0%*	*32.6%*
Gross Profit	**493,395**	**547,369**	**588,543**	**651,987**	**751,295**	**848,325**	**913,419**	**963,018**	**1,175,111**	**1,624,354**	**2,281,295**	**3,476,058**	**4,401,351**	**5,430,454**
Gross Margin	58.8%	58.6%	59.5%	62.3%	66.2%	67.0%	66.6%	66.8%	54.4%	56.1%	59.9%	66.7%	67.0%	67.4%
Total Opex	**(86,314)**	**(83,946)**	**(87,858)**	**(87,084)**	**(92,329)**	**(96,667)**	**(102,898)**	**(107,355)**	**(253,645)**	**(302,301)**	**(345,202)**	**(399,249)**	**(550,754)**	**(657,801)**
Percent of Revenues	*10.3%*	*9.0%*	*8.9%*	*8.3%*	*8.1%*	*7.6%*	*7.5%*	*7.5%*	*11.7%*	*10.4%*	*9.1%*	*7.7%*	*8.4%*	*8.2%*
R&D	(56,547)	(61,280)	(63,742)	(64,858)	(67,757)	(69,857)	(73,857)	(76,857)	(182,370)	(204,182)	(246,427)	(288,328)	(356,300)	(416,000)
Percent of Revenues	*6.7%*	*6.6%*	*6.4%*	*6.2%*	*6.0%*	*5.5%*	*5.4%*	*5.3%*	*8.4%*	*7.1%*	*6.5%*	*5.5%*	*5.4%*	*5.2%*
SG&A	(29,767)	(22,666)	(24,116)	(22,227)	(24,572)	(26,810)	(29,041)	(30,498)	(71,275)	(98,119)	(98,775)	(110,921)	(194,454)	(241,801)
Sequential Change	*1.1%*	*-23.9%*	*6.4%*	*-7.8%*	*10.6%*	*9.1%*	*8.3%*	*5.0%*						
Change vs Year Ago	*53.6%*	*-3.6%*	*-6.5%*	*-24.5%*	*-17.5%*	*18.3%*	*20.4%*	*37.2%*						
Operating Income	**407,081**	**463,424**	**500,685**	**564,902**	**658,966**	**751,658**	**810,521**	**855,664**	**921,466**	**1,322,053**	**1,936,092**	**3,076,809**	**3,850,597**	**4,772,653**
Percent of Revenues	*48.5%*	*49.6%*	*50.6%*	*54.0%*	*58.1%*	*59.3%*	*59.1%*	*59.4%*	*42.6%*	*45.7%*	*50.8%*	*59.0%*	*58.6%*	*59.2%*
Non-operating Income(Loss)	**23,815**	**29,612**	**24,684**	**27,461**	**28,834**	**26,507**	**26,507**	**26,507**	**57,706**	**83,786**	**105,571**	**108,356**	**106,030**	**102,030**
Profit Before Taxes	**430,895**	**493,035**	**525,369**	**592,363**	**687,800**	**778,165**	**837,029**	**882,171**	**979,171**	**1,405,839**	**2,041,663**	**3,185,165**	**3,956,627**	**4,874,683**
Percent of Revenues	*51.3%*	*52.8%*	*53.1%*	*56.6%*	*60.6%*	*61.4%*	*61.0%*	*61.2%*	*45.3%*	*48.6%*	*53.6%*	*61.1%*	*60.3%*	*60.5%*
Taxes	(70,163)	(95,542)	(73,614)	(86,948)	(114,999)	(132,288)	(125,554)	(132,326)	(141,404)	(233,407)	(326,266)	(505,167)	(612,395)	(766,866)
Tax Rate	*16.3%*	*19.4%*	*14.0%*	*14.7%*	*16.7%*	*17.0%*	*15.0%*	*15.0%*	*14.4%*	*16.6%*	*16.0%*	*15.9%*	*15.5%*	*15.7%*
Net Income, Cont Ops	360,733	397,493	451,755	505,415	572,801	645,877	711,474	749,846	837,768	1,172,432	1,715,397	2,679,998	3,344,231	4,107,817
Minority Interest	831	780	546	329	(321)	0	0	0	730	836	2,486	(321)	0	0
Reported Income (TW GAAP)	**361,564**	**398,273**	**452,301**	**505,744**	**572,480**	**645,877**	**711,474**	**749,846**	**838,498**	**1,173,268**	**1,717,883**	**2,679,677**	**3,344,231**	**4,107,817**
Percent of Revenues	*43.1%*	*42.7%*	*45.7%*	*48.3%*	*50.5%*	*51.0%*	*51.9%*	*52.0%*	*38.8%*	*40.5%*	*45.1%*	*51.4%*	*50.9%*	*51.0%*
Change vs Year Ago	*60.3%*	*60.7%*	*39.1%*	*35.0%*	*58.3%*	*62.2%*	*57.3%*	*48.3%*	*-17.5%*	*39.9%*	*46.4%*	*56.0%*	*24.8%*	*22.8%*
Reported EPS (NT$, TW GAAP)	**13.94**	**15.36**	**17.44**	**19.50**	**22.08**	**24.91**	**27.44**	**28.92**	**32.34**	**45.25**	**66.25**	**103.34**	**128.97**	**158.42**
Basic shares (mn units)	25,927	25,928	25,929	25,926	25,926	25,926	25,926	25,926	25,929	25,928	25,928	25,926	25,926	25,926

Source: Company data, Morgan Stanley Research (e) estimates

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Risk Reward – TSMC (2330.TW) Top Pick

2026 revenue and capex guidance raised

PRICE TARGET NT$2,588.00

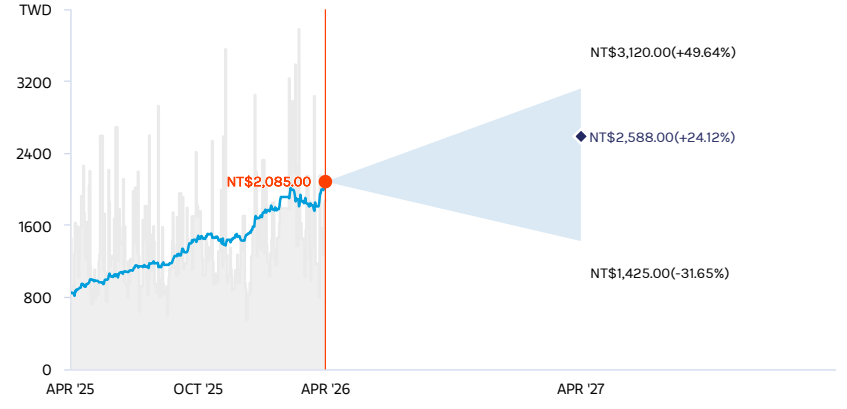
Base case, residual income model. Key assumptions: a cost of equity of 9.2% (beta of 1.2, risk-free rate of 2.0% and risk premium of 6.0%), an intermediate growth rate of 10.5%, and a terminal growth rate of 4.0%.

Consensus Price Target Distribution NT$1,267.00 NT$2,288.58 NT$3,030.00 MS PT

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Reflecting robust AI capex guidance from Meta and Microsoft, TSMC remains our Top Pick.
- Our price target reflects TSMC's strong profitability and another year of >30% Y/Y revenue growth seen for 2026e.
- Part of that can be contributed by China's AI GPU market through Nvidia.
- Reflecting increased bargaining power, long-term margin expansion, and sustainable AI semi demand, we expect the stock to re-rate to our implied 2027e target P/E of 20x.

Consensus Rating Distribution

97%	Overweight
3%	Equal-weight
0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era:	*Positive*
Pricing Power:	*Positive*
Secular Growth:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$3,120.00

24x 2027e EPS

TSMC dominates foundry services: 1) Breakthrough in EUV tech and materials accelerates node migration to 3nm and beyond. 2) Intel outsources its server CPU production to TSMC sooner than expected. 3) Intel or Samsung exits leading-edge foundry business. 4) New tech megatrends such as 6G or AI drive global semi revenue growth and increase demand for leading edge nodes. 5) Cost per transistor decreases further along with node migration.

BASE CASE NT$2,588.00

20x 2027e EPS

Leading position in leading-edge logic foundry services with some gross margin expansion: 1) Global semi revenue CAGR is sustained in the high single digits in 2025-28. 2) TSMC continues to lead in market share in 16nm, 7nm, 5nm, 3nm, and 2nm. 3) It continues to observe robust AI-related demand. 4) Cost per transistor does not decrease meaningfully in 3nm and 2nm, while capex per k is more demanding than for 7nm and 5nm.

BEAR CASE NT$1,425.00

11x 2027e EPS

TSMC's dominance in foundry business erodes: 1) Global ex-memory semi revenue growth is weaker than expected in 2025-28. 2) Declining trailing-edge market share causes revenue shortfall. 3) Intel's contribution is weaker than expected. Intel and/or Samsung successfully develop leading-edge process technology. 4) Intel's foundry business execution proves successful and weighs on TSMC with order competition. 5) Demand for 2nm is less than expected given high transistor cost for foundry customers.

Risk Reward – TSMC (2330.TW)

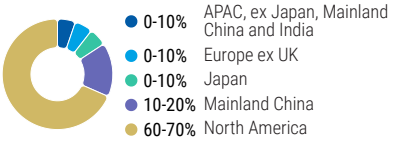
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue from 5nm Geometry (NT$, mn)	1,179,907	1,224,675	1,020,463	1,051,528
Revenue from 10/7nm Geometry (NT$, mn)	459,258	501,375	496,279	494,427
Revenue from 16/20nm Geometry (NT$, mn)	220,027	289,383	318,988	336,404

INVESTMENT DRIVERS

- Moore's Law migration (chip scaling) and alternative technologies
- Capex intensity - cost of building leading-edge foundry capacity
- Bargaining power with key equipment vendors
- Transistor cost for customers
- ROI for leading-edge foundry

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 **MOST** 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- TSMC charges large customers more to keep its gross margin above 56% in the long term.
- AI semi demand grows more significantly than expected, while TSMC maintains high market share in leading-edge foundry business.
- Outsourcing from Intel CPU increases in 2025-27.

RISKS TO DOWNSIDE

- Inventory correction occurs in 2026.
- Demand for leading edge technologies weakens.
- Costs of overseas fabs grow significantly.

OWNERSHIP POSITIONING

Inst. Owners, % Active 72.8%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	4,456,000	5,213,620 / 5,023,426	5,266,869
EBITDA (NT$, mn)	3,062,907	3,589,886	3,882,631
Net income (NT$, mn)	2,005,000	2,370,158	2,679,677
EPS (NT$)	77.30	91.03	103.34

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial summary

Income Statement, 2024-2028e, Year End Dec

(NT$ mn)	2024	2025	2026e	2027e	2028e
Revenue	**2,894,308**	**3,809,054**	**5,213,620**	**6,566,633**	**8,060,022**
YoY Growth	*33.9%*	*31.6%*	*36.9%*	*26.0%*	*22.7%*
Gross Profit b/f dep	2,247,086	2,924,670	4,212,960	5,364,278	6,756,268
Less: COGS	**(1,269,954)**	**(1,527,760)**	**(1,737,562)**	**(2,165,282)**	**(2,629,568)**
Variable costs	(647,221)	(884,384)	(1,000,659)	(1,202,355)	(1,303,754)
Depreciation & amort	(622,733)	(643,376)	(736,902)	(962,927)	(1,325,814)
Gross profit	**1,624,354**	**2,281,295**	**3,476,058**	**4,401,351**	**5,430,454**
% margin	*56.1%*	*59.9%*	*66.7%*	*67.0%*	*67.4%*
YoY Growth	*38.2%*	*40.4%*	*52.4%*	*26.6%*	*23.4%*
Operating Expenses:	**(302,301)**	**(345,202)**	**(399,249)**	**(550,754)**	**(657,801)**
R&D	(204,182)	(246,427)	(288,328)	(356,300)	(416,000)
Sales and Marketing	(13,144)	(16,918)	(26,068)	(32,833)	(40,300)
General and Admin	(84,975)	(81,857)	(84,853)	(161,621)	(201,501)
Operating Profit	**1,322,053**	**1,936,092**	**3,076,809**	**3,850,597**	**4,772,653**
% margin	*45.7%*	*50.8%*	*59.0%*	*58.6%*	*59.2%*
YoY Growth	*43.5%*	*46.4%*	*58.9%*	*25.1%*	*23.9%*
Non-Operating Income	**83,786**	**105,571**	**108,356**	**106,030**	**102,030**
Pretax Profit	**1,405,839**	**2,041,663**	**3,185,165**	**3,956,627**	**4,874,683**
% margin	*48.6%*	*53.6%*	*61.1%*	*60.3%*	*60.5%*
Tax	(233,407)	(326,266)	(505,167)	(612,395)	(766,866)
Reported net Income	**1,173,268**	**1,717,883**	**2,679,677**	**3,344,231**	**4,107,817**
Reported EPS (NT$)	**45.25**	**66.25**	**103.34**	**128.97**	**158.42**
EPS for consensus (NT$)	**45.25**	**66.25**	**103.34**	**128.97**	**158.42**

Key Ratios, 2024-2028e

	2024	2025	2026e	2027e	2028e
Return (%)					
ROAA	19%	23%	30%	29%	29%
ROAE	30%	35%	41%	38%	35%
OP. ATO	0.7x	0.7x	0.8x	1.0x	0.9x
Gearing (x)					
Net Debt/ Equity	-0.33x	-0.38x	-0.41x	-0.43x	-0.48x
Current Ratio	2.4x	2.6x	3.4x	4.3x	5.7x
Quick Ratio	1.9x	2.1x	2.8x	3.7x	5.1x
Operating Cycle					
AR/NR Turnover (days)	29	26	24	25	25
Inventory Turnover (days)	76	68	66	65	65
AP Turnover (days)	19	19	19	19	19
Cash Conversion (days)	87	75	70	71	70

e = Morgan Stanley Research Estimates
Source: Company Data, Morgan Stanley Research

Balance Sheet, 2024-2028e, Year End Dec

(NT$ mn)	2024	2025	2026e	2027e	2028e
Cash & Equivalent	2,127,627	2,767,856	3,791,590	5,064,922	7,098,277
MarketableSecurity	294,393	300,738	300,738	300,738	300,738
A/R & N/R	272,088	281,791	404,939	509,873	612,439
Inventories	287,869	288,109	348,714	428,868	513,486
Other Current Ass.	106,376	178,635	178,635	178,635	178,635
Total current assets	**3,088,352**	**3,817,131**	**5,024,617**	**6,483,037**	**8,703,575**
Long-term Investment	149,040	172,370	172,370	172,370	172,370
Total fixed assets	3,234,980	3,691,841	4,612,679	5,770,237	6,723,126
Total other assets	219,565	251,682	251,682	251,682	251,682
Total Assets	**6,691,938**	**7,933,024**	**10,061,349**	**12,677,325**	**15,850,753**
A/P & N/P	74,227	84,330	103,493	127,282	152,395
Accrued Expenses	0	0	0	0	0
Other Payable	531,377	630,217	630,217	630,217	630,217
Curr. of L-T Debt	59,858	136,926	136,926	136,926	136,926
Other Curr. Liab.	599,064	606,547	606,547	606,547	606,547
Total Current Liab.	**1,264,525**	**1,458,019**	**1,477,182**	**1,500,971**	**1,526,084**
L-T Liabilities	926,605	856,228	856,228	856,228	856,228
Total Other L-T Liab	177,233	157,982	157,982	157,982	157,982
Total Liabilities	**2,368,362**	**2,472,229**	**2,491,392**	**2,515,180**	**2,540,294**
Common Stocks	259,327	259,325	259,325	259,325	259,325
Preferred Stocks	0	0	0	0	0
Capital Reserve	73,261	73,446	73,446	73,446	73,446
Retained earnings	3,955,957	5,086,825	7,195,987	9,788,175	12,936,489
Treasury Stock	0	0	0	0	0
Minority Equity	35,031	41,199	41,199	41,199	41,199
Total Equity	**4,323,576**	**5,460,795**	**7,569,957**	**10,162,145**	**13,310,459**
Total Liab. & Equity	**6,691,938**	**7,933,024**	**10,061,349**	**12,677,325**	**15,850,753**

Cash Flow Statement, 2024-2028e, Year End Dec

(NT$ mn)	2024	2025	2026e	2027e	2028e
Net Income Current	1,173,268	1,717,883	2,679,677	3,344,231	4,107,817
Depreciation	662,797	688,096	805,822	1,013,607	1,395,594
Net Investment Losses (Gains)	8,205	4,992	0	0	0
Decrease (Increase) in Working Capital	20,896	(17,144)	(164,589)	(161,299)	(162,070)
Others	(38,988)	(118,851)	0	0	0
Cash Flow-Operating	**1,826,177**	**2,274,976**	**3,320,910**	**4,196,540**	**5,341,341**
(Purchase) of FA	(956,007)	(1,272,411)	(1,726,660)	(2,171,164)	(2,348,483)
Sale of Fix Asset	895	797	0	0	0
(Purchase)L-T Inv.	(3,739)	0	0	0	0
Others	94,008	127,220	0	0	0
Cash Flow-Inv.	**(864,843)**	**(1,144,393)**	**(1,726,660)**	**(2,171,164)**	**(2,348,483)**
Inc(Dec)-S-T Debt	0	0	0	0	0
Inc(Dec) L-T Debt	28,601	7,948	0	0	0
Dividend Paid	(363,055)	(466,779)	(570,516)	(752,043)	(959,503)
Dir.&Emp.Bonus	0	0	0	0	0
Others	(11,847)	18,486	0	0	0
Cash Flow-Financing	**(346,301)**	**(440,345)**	**(570,516)**	**(752,043)**	**(959,503)**
Change in Cash	**662,199**	**640,229**	**1,023,734**	**1,273,332**	**2,033,354**
Net cash/(debt), b/f	1,465,428	2,127,627	2,767,856	3,791,590	5,064,922
Net cash/(debt), c/f	**2,127,627**	**2,767,856**	**3,791,590**	**5,064,922**	**7,098,277**

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UMC (2303.TW)

Base-case, residual income model. Key assumptions: 9.2% cost of equity, 5.9% medium-term growth, and 3.0% terminal growth.

Risks to Upside

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- Stronger-than-expected global economic and semiconductor growth
- Looser pricing competition, increasing margins and utilization rates
- Favorable customer mix changes

Risks to Downside

- Softer-than-expected global economic and semiconductor growth
- Increasing pricing competition, reducing margins and utilization rates
- Unfavorable customer mix changes

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